WORKED SOLUTION

# STEP II Specimen paper, Question 16

Probability and Statistics • Tree diagram

**STEP II Specimen paper, Question 16**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q16

This question has a full guided treatment in *Beyond STEP*, Section 12.7, *Who Meets Whom in a Knockout*: https://beyondstep.keithmcnulty.org/ch12/knockout.html

## The key idea

Follow the two chosen players, $A$ and $B$, round by round. In any round where both are still in and have not yet met, only two things matter: is $A$ drawn against $B$, and if not, do both of them win? Each of these is a simple probability, and the products that appear telescope beautifully. For the last part there is also a slick symmetry argument: count the matches in the whole tournament.

## Solution

Write $N = 2^n$. Call the two chosen players $A$ and $B$. After $r-1$ rounds there are $2^{n-r+1}$ players left, so round $r$ has $M_r = 2^{n-r+1}$ players, and round $n$ (with $M_n = 2$) is the final.

The fact we use repeatedly: if both $A$ and $B$ are among $M$ players who are paired at random, then $A$'s opponent is equally likely to be any of the other $M-1$ players, so

$$\mathrm{P}(A \text{ is drawn against } B) = \frac{1}{M-1}.$$

(In the final there is no draw, but with $M = 2$ the formula gives 1, which is right.)

### Part (i)

In the first round $M_1 = N$, so

$$\mathrm{P}(A \text{ and } B \text{ meet in the first round}) = \frac{1}{2^n - 1}.$$

**Part (ii)**

Let $S_r$ be the event that $A$ and $B$ are both still in the tournament at the start of round $r$ (which forces them not to have met yet, since the loser of a match is out). Given $S_r$, they reach round $r+1$ together exactly when they are *not* drawn together in round $r$ and then both win their (separate) matches. The two match results are independent, each with probability $\frac{1}{2}$, so

$$\mathrm{P}(S_{r+1} \mid S_r) = \frac{M_r - 2}{M_r - 1} \cdot \frac{1}{4}.$$

**Ask yourself** How do we turn this into a product that we can actually evaluate?

Write $M = 2^k$, and notice that $M - 2 = 2(2^{k-1} - 1)$, so

$$\frac{M-2}{M-1} = \frac{2\,(2^{k-1}-1)}{2^k - 1}.$$

The numerator for one round is (apart from the factor 2) the denominator for the next round. Since $S_1$ is certain, multiplying over the rounds with $k = n, n-1, \ldots, n-r+2$ gives a telescoping product:

$$\mathrm{P}(S_r) = \frac{1}{4^{r-1}} \prod_{k=n-r+2}^{n} \frac{2\,(2^{k-1}-1)}{2^k - 1} = \frac{2^{r-1}}{4^{r-1}} \cdot \frac{2^{n-r+1}-1}{2^n - 1} = \frac{2^{n-r+1}-1}{2^{r-1}\,(2^n - 1)}.$$

Given $S_r$, they meet in round $r$ with probability $1/(M_r - 1) = 1/(2^{n-r+1} - 1)$, so

$$\mathrm{P}(A \text{ and } B \text{ meet in round } r) = \frac{1}{2^{r-1}\,(2^n - 1)}.$$

This is a pleasantly simple formula: each round halves the chance of meeting there. Setting $r = 1$ recovers part (i). For the final, $r = n$:

$$\mathrm{P}(A \text{ and } B \text{ meet in the final}) = \frac{1}{2^{n-1}\,(2^n - 1)}.$$

As a sanity check by symmetry: the rules treat every player identically, so the pair of finalists is equally likely to be any of the $\binom{N}{2} = 2^{n-1}(2^n - 1)$ pairs of players, which gives the same answer.

**Part (iii)**

The events "meet in round $r$" for $r = 1, \ldots, n$ are mutually exclusive (two players can meet at most once, as the loser leaves), so we add, using the geometric series:

$$\begin{aligned}\mathrm{P}(A \text{ and } B \text{ meet}) &= \frac{1}{2^n - 1} \sum_{r=1}^{n} \frac{1}{2^{r-1}} = \frac{1}{2^n - 1} \cdot 2\left(1 - \frac{1}{2^n}\right) \\ &= \frac{2}{2^n} \cdot \frac{2^n - 1}{2^n - 1} = \frac{1}{2^{n-1}}.\end{aligned}$$

**Ask yourself** Is there a way to see this answer without any summing?

Yes. Every match eliminates exactly one player and all but the champion are eliminated, so there are always exactly $2^n - 1$ matches. Each match is played by a pair of players, and by symmetry every one of the $\binom{N}{2}$ pairs has the same probability $p$ of playing. The expected number of matches is the sum of these probabilities over all pairs, so

$$\binom{2^n}{2} p = 2^n - 1 \quad \Longrightarrow \quad p = \frac{2(2^n - 1)}{2^n(2^n - 1)} = \frac{2}{2^n} = \frac{1}{2^{n-1}},$$

in agreement with the sum.

## Checks

- $n = 1$: two players, who certainly meet in the first round, which is the final. All three formulas give 1.
- $n = 2$: four players. Part (i) gives $\frac{1}{3}$, the final gives $\frac{1}{6}$, and the total is $\frac{1}{3} + \frac{1}{6} = \frac{1}{2}$, as part (iii) says.
- $n = 3$: enumerating every draw and every result exactly gives $\frac{1}{7}$, $\frac{1}{28}$ and $\frac{1}{4}$. A simulation of $2 \times 10^5$ tournaments with 16 players gave 0.0664, 0.0085 and 0.1247, against the exact $\frac{1}{15} \approx 0.0667$, $\frac{1}{120} \approx 0.0083$ and $\frac{1}{8}$.

## Common slips

- Forgetting that, to meet in a later round, the two players must first *avoid* each other in every earlier round: the factor $\frac{M-2}{M-1}$ is easy to drop.
- Answering part (ii) with $\left(\frac{1}{2^{n-1}}\right)^2$, the chance that both reach the final *if they were independent*. They are not: if they are drawn together early, at most one of them can get through.

## Beyond STEP

The argument in part (iii) is *linearity of expectation* with indicator variables: the expected number of matches is the sum, over all pairs, of the probability that the pair plays, whether or not the events are independent. This is one of the most powerful tools in probability and combinatorics, and it underlies the "probabilistic method" for proving that objects with given properties exist. Blitzstein and Hwang, *Introduction to Probability*, has a full chapter of lovely examples of indicator-variable arguments.

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